HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · POWER TRANSMISSION · PROJECT PLANNING

TRANSMISSION LINE PACKAGES · 14 STRETCHES · 3 STAGES

Schedule Automation

A VBA schedule engine inside Excel: a field delay in foundation, erection or stringing now ripples through every downstream stretch in seconds, and planners see the result on a Gantt chart.

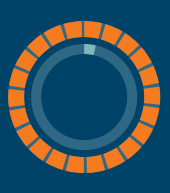
~~12h~~ → **2 min** to re-forecast after a field change



2 min
Time to re-forecast after a field change
▼ from about 12 hours of edits



168 levers
Schedule levers in one form
14 stretches × 3 stages × 4 levers



22 a month
Re-forecasts per month
▲ from one monthly update

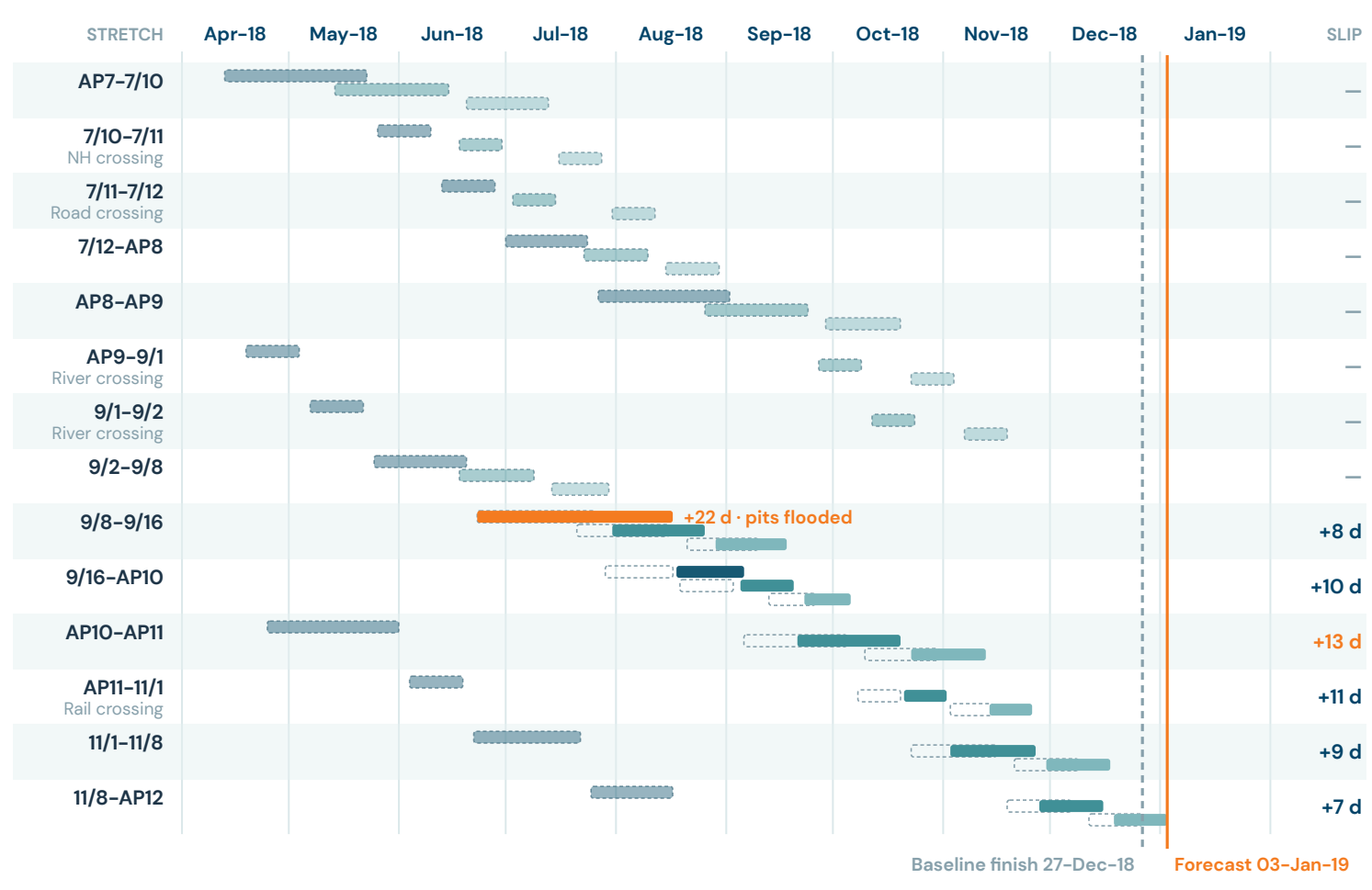


3 weeks
Earlier warning of finish-date slippage
seen while still recoverable

Rings: a 12-hour re-plan with the 2-minute re-forecast as the orange sliver inside; 14 stretches around 3 stages; 22 re-forecasts against the old single monthly update; 3 weeks of earlier warning against the 6-week slip the old process surfaced late. Figures indicative; validated metrics available on request.

01 One flooded stretch, the whole line re-plans

Sample 14-stretch package · baseline vs forecast after one field event



FIELD EVENT

Monsoon water-logging

Foundation pits at 9/8-9/16 flooded: +22 days. One entry in the form.

RULES THE ENGINE ITERATES

Stages and gangs are linked

Erection starts once 55% of foundations are done plus 7 days of curing; stringing starts 3 days after a stretch is fully erected; each gang works its stretches in order.

RESULT, RECALCULATED

Every downstream date settles

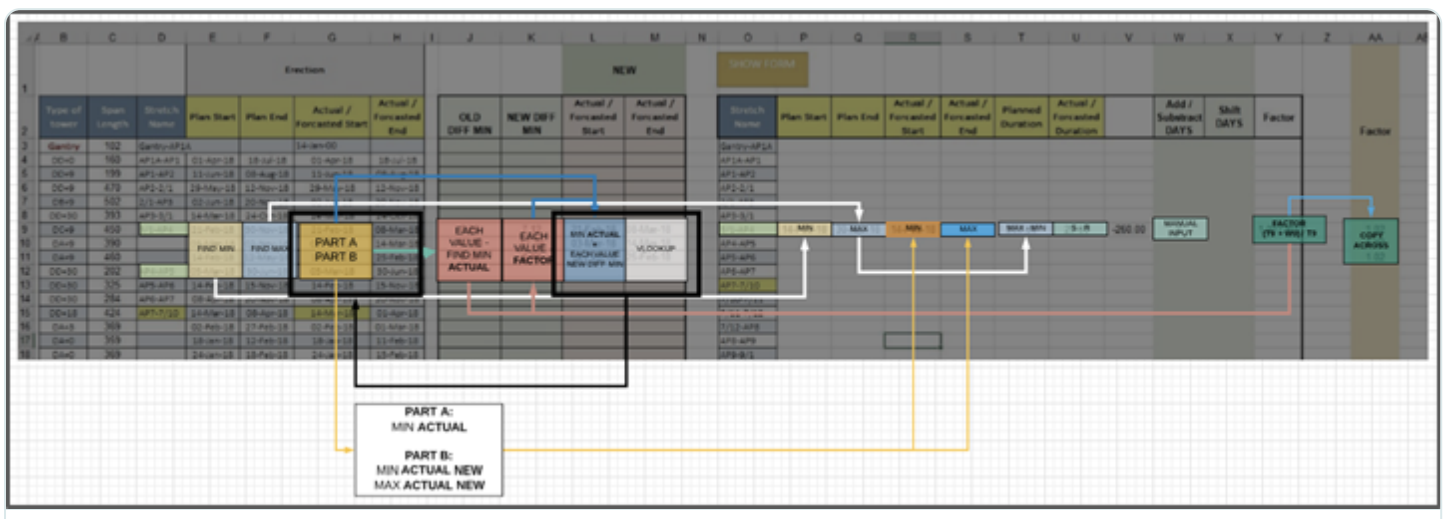
Six stretches move, down to the last stringing gang. The finish date shifts by a week, and planners see it the same day.

+7 days at completion
14 of 42 bars moved

Sample package for illustration: stretch names follow the original form; durations, gang counts and lags are indicative.

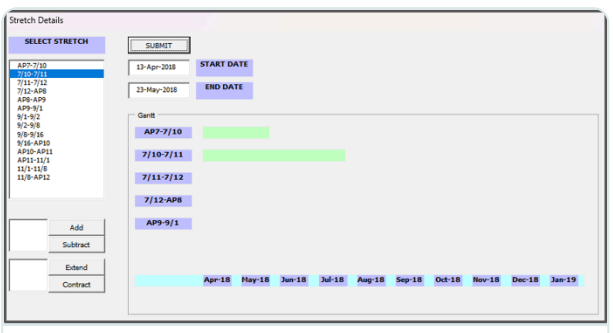
02 The workbook, as built

Original screens, shown as delivered



1 **Base sheet**: tower type, span length and stretch, then plan and actual/forecast dates per stage. Coloured boxes show how each date is recalculated.

1. **FIND MIN / MAX** of the stretch's plan dates
2. **PART A** = min actual; **PART B** = min and max actual new
3. **Each value** rescaled from min actual using the factor
4. **VLOOKUP** brings in the stretch's new start
5. **MIN, MAX, MAX-MIN** give planned and forecast durations
6. **FACTOR = (T6 + R6) / T6** from the manual add/subtract days



2 **Stretch Details**: pick a stretch such as 7/10-7/11, read its start and end, adjust it and see its Gantt chart.



3 **USER CONTROL**: pick base data, list stretches, choose foundation, erection or stringing, then add or shift days. Refresh From Base resets the plan.

RESCALING RULE

New date = new minimum + (old date - old minimum) × factor, where factor = (planned duration + days added) ÷ planned duration.

03 Four pain points, four fixes

Before → after

BEFORE
Two tools, one field
Baseline in Microsoft Project, time–location view in TILOS; every update re-keyed twice.

ONE EXCEL ENGINE

AFTER
Loaded once from the baseline
Project and TILOS stay as inputs; daily re-planning happens where planners already work.

BEFORE
Three stages, one chain
One late foundation pushes erection and stringing downstream, traced by hand stage by stage.

VBA ITERATIONS

AFTER
The cascade runs itself
VBA iterates through every stretch and stage until every downstream date settles.

BEFORE
Too many levers
Start, duration, shift and factor at each stage: 168 linked levers in a 14-stretch package.

USERFORMS

AFTER
One form, every lever
Pick a stretch and stage, add or shift days; the Gantt chart refreshes on submit.

BEFORE
Re-plans arrived late
Monsoon, right-of-way and crossings changed the plan weekly; a re-plan took about a day.

DAILY RE-FORECAST

AFTER
Test, see, then decide
About 2 minutes per change; what-ifs are cheap, and Refresh From Base keeps the baseline.

04 Hours to re-plan

One field change, before (light) and after (navy) · indicative

12 h → 29 min per field change, capture to circulation

05 Slip seen in time

Milestone trend: flat is good, late jumps are bad

06 How HMC ran it

1 week	**Discover** Mapped every lever and dependency; timed a typical re-plan at about 12 hours.
1 week	**Design** One stretch-wise base sheet per stage, filled from the Project baseline.
3 weeks	**Build** VBA iteration engine with USER CONTROL and Stretch Details forms.
1 month	**Adopt** Site delays reach the schedule the same day; weekly forecast-vs-baseline review.
∞ ongoing	**Scale** Template for every line package; forecasts fed back to the master schedule.

What any project team can borrow

01
Model the levers, not the dates
Once start, duration, shift and factor are inputs, every date becomes a result. Planners change one lever instead of fifty cells.

02
Let the iterations carry the cascade
Dependencies between stages and gangs are rules, not memory. The engine runs them until the plan settles, every time.

03
Meet planners in Excel
A UserForm on the sheet they already trust was adopted in weeks. New software would have taken a project cycle.

Built with **Excel** · **VBA** · **VBA UserForms** · **Excel Gantt charts**, fed by **Microsoft Project** · **TILOS** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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